

1D Ising continued - spatial correlations

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$$G(i, j) = \langle s_i s_j \rangle \quad (1)$$

Computation of $G(i, j)$

→ $H = 0$ - magnetic field

→ $j = i + 1$ first compute $G(i, i + 1)$

$$G(i, i + 1) = \sum_{s_1, \dots, s_N} e^{\beta J s_1 s_2 + \dots + \beta J_i s_i s_{i+1} + \dots + \beta J_{N-1} s_{N-1} s_N} = \frac{\partial}{\partial (\beta J)} \ln(Z(J_i)) \quad (2)$$

We can compute the partition function for partition functions for different couplings?

$$Z(J_I) = 2^N \prod_{i=1}^{N-1} \cosh(\beta J_i) \quad (3)$$

$$G(i, i + 1) = \langle s_i s_{i+1} \rangle = \tanh(\beta J_i) \rightarrow \tanh(\beta J) \quad (4)$$

Can we compute $G(i, j)$

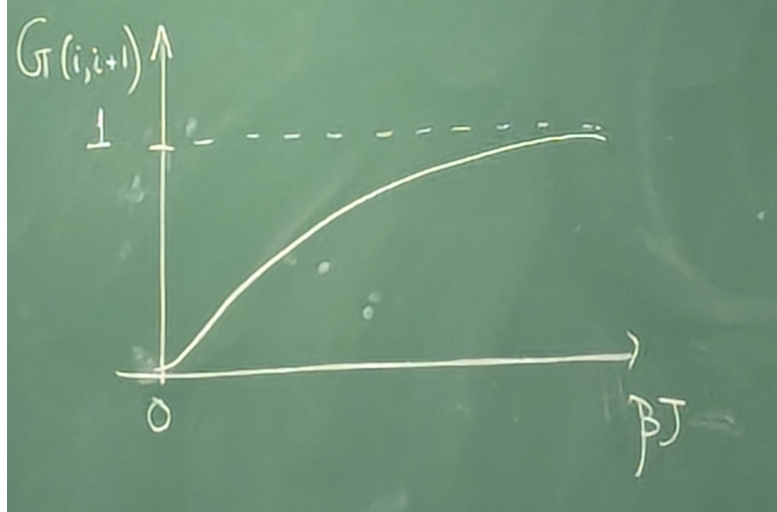


Figure 1: The probability touches 1 at zero temperature

Consider the case $j = i + 2$

We can sue the fact that $s_k^2 = 1$

$$s_i s_{i+2} = s_i s_{i+1}^2 s_{i+2} \implies G(i, i + 2) = \langle (s_i s_{i+1}) (s_{i+1} s_{i+2}) \rangle = \frac{\partial}{\partial (\beta J_i)} \frac{\partial}{\partial (\beta J_{i+1})} \ln Z \quad (5)$$

Differentiating wrt to i and $(i+1)$ We get

$$G(i, i + 2) = \tanh(\beta J_i) \tanh(\beta J_{i+1}) \rightarrow_{J_i \equiv J} \tanh^2(\beta J) \quad (6)$$

$$G(i, i + j) = \frac{\partial}{\partial (\beta J_i)} \frac{\partial}{\partial (\beta J_{i+1})} \dots \frac{\partial}{\partial (\beta J_{i+j-1})} \ln Z(J_k) \quad (7)$$

We get prod of tanh's but if we let $J \equiv J_i \forall i$ then we get a power.

$$\approx \tanh(\beta J_i) \dots \tanh(\beta J_{i+j-1}) \quad (8)$$

$$\implies G(i, i + j) = \tanh^j(\beta J) \quad (9)$$

which only holds if all J are equal.

Feature we will see in QFT for Poincare invariant.

Here we have discrete transformations and this result is translationally invariant. The result only depends on the distance j between the lattice sites and for continuous variables e.g. in QFT $G(r, r') = G(|r - r'|)$

What can we learn from $G(i, i + j)$?

For $H = 0$, when $\beta J \gg 1$

$$G(i, i + j) = \tanh^j(\beta J) \quad (10)$$

For $H = 0$, when $\beta J \gg 1$

Then all the spins are up or all the spins are down.

We see a 'long range order'.

We will define a physical quantity when there is or is not long range order.

Let us express $G(i, i + j)$

$$= \tanh^j(\beta J) = \coth^{-j}(\beta J) = e^{-j \ln[\coth(\beta J)]} \equiv e^{-\frac{j}{l}} \quad (11)$$

define

$$\frac{1}{l} \equiv \ln[\coth(\beta J)] \quad (12)$$

$$l \equiv \frac{1}{\ln[\coth(\beta J)]} \equiv \text{correlation length for system} \quad (13)$$

If we introduce a magnetic field it should depend on the magnetic field.

We see that correlations decay exponentially as soon as (βJ) diminishes or as we move away from $T = 0$

For $\beta J \rightarrow \infty$, $l \rightarrow \infty$

For $\beta J \rightarrow 0$, $l \rightarrow 0$

Divergence of the correlation length indicates a phase transition. How it diverges characterises the type of phase transition.

How does l diverge as we approach $T \rightarrow 0$?

i.e. expand and look at leading order behavior.

$$\coth(\beta J) = \frac{1 + e^{-\beta J}}{1 - e^{-\beta J}} = \frac{1 - e^{2\beta J} + 2e^{-2\beta J}}{1 - e^{-2\beta J}} \approx 1 + 2e^{-2\beta J} + \dots \quad (14)$$

here the approximation is under $\beta J \gg 1$

$$l \approx \frac{1}{2} e^{2\beta J} \quad (15)$$

diverges exponentially! 'essential singularity'

For $d \geq 2$ where p.t. continuous $l \approx (T - T_c)^{-\nu}$ where the approximation is as $T \rightarrow T_c$.

How to compute the correlation function for $H \neq 0$.

First let us compute $\langle s_i \rangle$ ('when no mag field its zero')

$$\langle s_i \rangle = \frac{1}{Z} \sum_{s_1, \dots, s_N} s_i e^{-\beta H} \quad (16)$$

$$H = -J \sum s_i s_{i+1} - H \sum s_i \quad (17)$$

We recall we had that we can write exponential a product of matrix elements

$$\langle s_i \rangle = \frac{1}{Z} \sum_{s_1, \dots, s_N} s_i \mathcal{T}_{s_1 s_2} \dots \mathcal{T}_{s_{i-1} s_i} \mathcal{T}_{s_i s_{i+1}} \dots \mathcal{T}_{s_{N-1} s_N} \quad (18)$$

We group terms

$$= \sum_{s_1} \sum_{s_2} \dots \sum_{s_i} \dots \sum_{s_N} \mathcal{T}_{s_1 s_2} \dots \mathcal{T}_{s_{i-1} s_i} s_i \mathcal{T}_{s_i s_{i+1}} \dots \mathcal{T}_{s_N s_1} \quad (19)$$

What happens to $\mathcal{T}_{s_{i-1} s_i} s_i \mathcal{T}_{s_i s_{i+1}}$
consider

$$\sum_{s_i} \mathcal{T}_{s_{i-1} s_i} s_i \mathcal{T}_{s_i s_{i+1}} = \mathcal{T} \cdot \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \cdot \mathcal{T} \equiv \mathcal{T} \sigma_z \mathcal{T} \quad (20)$$

where we have the third pauli matrix

What this means is

$$\langle s_i \rangle = \frac{1}{Z} \text{Tr} \left(\sigma_z \mathcal{T}^N \right) \quad (21)$$

We need to diagonal's to find $\mathcal{T}^N$. We cant perform the same trick as before as we now have the pauli matrix here.

If $\mathcal{T}$ is diagonalised by Λ

$$\text{Tr} \left(\Lambda^{-1} \sigma_z \Lambda \Lambda^{-1} \mathcal{T}^N \Lambda \right) = \text{Tr} \left(\Lambda^{-1} \sigma_z \Lambda \mathcal{T}_D^N \right) \quad (22)$$

If doing exerises we can let

$$\Lambda \equiv \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad (23)$$

What about

$$G_H(i, i+j) = \frac{1}{Z} \sum_{s_1, s_2, \dots, s_N} e^{-\beta H} s_i s_{i+j} \quad (24)$$

$$= \frac{1}{Z} \sum_{s_1} \dots \sum_{s_i} \dots \sum_{s_{i+j}} \dots \sum_{s_N} \mathcal{T}_{s_1 s_2} \dots \mathcal{T}_{s_{i-1} s_i} s_i \mathcal{T}_{s_i s_{i+1}} \dots \mathcal{T}_{s_{i+j-1} s_{i+j}} \dots \mathcal{T}_{s_N s_1} \quad (25)$$

$$= \frac{1}{Z} \text{Tr} \left(\sigma_z \mathcal{T}^j \sigma_z \mathcal{T}^{N-j} \right) \quad (26)$$

We get

$$G_H(i, i+j) \propto e^{-j \ln \left(\frac{\lambda_1}{\lambda_2} \right)} \quad (27)$$

in the thermo limit ie ($N \rightarrow \infty$ and $\lambda_1 > \lambda_2$)

$$l \propto \frac{1}{\ln \left(\frac{\lambda_1}{\lambda_2} \right)} \quad (28)$$

The phase transition we saw at $T = 0$ is an example of level crossing.

In general if we have a system that depends on level crossing we introduce a new variable and we say the levels don't change much. In the new system the change of level is such that one cross the other. What used to be an excited state is now the ground state. That is why there is a phase transition is at $T = 0$.